
PROJECT SYNOPSIS

EduFeedback AI System

Dynamic Student Feedback Management & AI-Powered Analytics Portal

Project Title:	EduFeedback AI Student Feedback System
Institution Context:	Dr. A.P.J. Abdul Kalam Technical University (AKTU)
Technology Stack:	FastAPI, React (Vite), SQLite, Hugging Face Transformers
Developer Profile:	Hemansh Verma (Student Roll: 2000290100001)
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1. Introduction

In modern academic settings, continuous feedback is essential to monitor teaching quality, course structures, and institutional facilities. Traditional course evaluations often rely on paper surveys or static online questionnaires that fail to capture detailed pedagogical insights. The **EduFeedback AI System** is designed as an intelligent web-based platform tailored for academic institutions like Dr. A.P.J. Abdul Kalam Technical University (AKTU) to digitize, analyze, and streamline this evaluation process.

By merging modern responsive interface patterns with natural language processing pipelines, the system transitions student feedback from binary ratings into actionable corrective strategies. Administrators can design evaluation questionnaires dynamically, students can evaluate instructors securely, and faculty members can review automated AI sentiment summaries without manual report compiling.

2. Problem Statement

Existing student feedback frameworks suffer from several critical shortcomings:

- **Deaf to Descriptive Context:** Traditional systems focus on star ratings, neglecting text descriptions. Critical details about pacing, lab hardware issues, and textbook constraints remain buried in comments because staff cannot read thousands of feedbacks manually.
- **Rigid Question Forms:** Academic metrics change yearly. Hardcoded web forms require software redeployments to add new questions or edit evaluation criteria, causing high maintenance overheads.
- **Static Average Reporting:** Standard reports calculate raw averages, which hide local sentiment. A teacher might score 4.0 but have strong negative complaints about classroom delivery speed that average ratings obscure.
- **Information Silos:** Staff cannot query feedback database systems interactively, resulting in lag times when attempting to retrieve targeted performance reports.

3. Project Objectives

The primary goals of the EduFeedback AI system are:

1. **Build a Secure Multi-Role Portal:** Establish dedicated flows for students, faculty members, and system administrators with unique security scopes.
2. **Implement a Dynamic Form Designer:** Equip administrators with visual tools to create, rearrange, edit, and publish questionnaire blocks in real-time.
3. **Automate Sentiment Analysis:** Integrate a local Hugging Face NLP pipeline to extract sentiment labels (Positive, Neutral, Negative) and keywords on descriptive comments.
4. **Deliver Corrective Recommendations:** Build intelligent rule-based systems mapping low ratings to corrective actions (e.g., advising slide reviews or lab updates).
5. **Create a Database Query Chatbot:** Provide a chatbot helper allowing administrators to search feedback trends using normal conversational queries.

4. Proposed Solution

The proposed solution is an end-to-end Single Page Application (SPA) backed by a REST API and a local machine learning inference engine. When a student submits a review, the system runs the text through an NLP pipeline, writes the feedback to a relational database, and re-calculates department metrics. The administrator has immediate access to an interactive analytics dashboard, a form designer to alter student questions, and a chatbot helper to search the logs.

5. Methodology

The project development follows a structured agile cycle divided into five primary stages:

1. **Requirement Definition & DB Modeling:** Design the relational database layout in SQLite with tables for Users, Subjects, Teachers, Feedbacks, and Questionnaire Templates.
2. **API Development (FastAPI):** Set up secure endpoints for token generation (JWT auth), student form submissions, and administrative designer updates.
3. **AI Inference Integration:** Set up Uvicorn to load Hugging Face pipelines (DistilBERT for sentiment) locally on startup to keep all feedback data private on university servers.
4. **Frosted Glassmorphic Frontend:** Code the frontend using Vite and React, applying a premium frosted glass aesthetic (`backdrop-filter`) and smooth transition animations.
5. **Verification & Optimization:** Conduct end-to-end validation of the questionnaire stepper, designer state propagation, and chatbot responses.

6. Main Modules and Features

A. Student Evaluation Module

Features a multi-step stepper form allowing students to select a subject and teacher, fill out dynamic rating questions, write observations, and review a final summary sheet before submitting.

B. Questionnaire Form Designer Workspace

An admin control panel containing template modification tools. Admins can rename section headers, append new evaluation metrics, delete sections, and publish templates to local storage to dynamically update the student portal.

C. Executive AI Analytics Dashboard

Renders key indicators: average satisfaction score, positive-to-negative ratios, keyword clouds, and critical negative mentions list. Renders automated faculty corrective recommendations based on satisfaction score averages.

D. Database Query Chatbot Assistant

An embedded conversational assistant using sentence similarity logic. It allows admins to type queries like *'Which subject has the lowest score?'* and receives instant database summaries.

E. Accounts Management Console

Provides user controls to add new student/faculty accounts, search roles, and block/unblock users from submitting evaluations.

7. Project Scope

In Scope:

- Real-time customization of student feedback forms.
- Automatic text parsing (sentiment + keywords) on student comments.
- Role-based access controls and account blockage triggers.
- Interactive database querying via conversational chatbot helper.

Out of Scope:

- Multi-tenant deployment (designed for single university system).
- Cloud database integration (uses SQLite for local execution).

8. Technology Stack

Layer	Technologies Used	Function
Frontend	React 18, Vite, TailwindCSS, Lucide React, Recharts	Frosted glassmorphism layout, multi-step stepper, designer workspace
Backend	Python 3, FastAPI, Uvicorn, SQLAlchemy	High performance RESTful APIs, JWT token authorization, route filters
Database	SQLite	Relational data storage, indexing, local development integrity
AI Engine	Hugging Face Hub (Transformers: distilbert-base-uncased)	Descriptive text sentiment analysis and NLP keyword mining

9. Expected Outcome

The completed platform delivers a centralized, smart feedback management portal. By analyzing student comments automatically, institutions can isolate course and faculty issues in seconds. The dynamic form designer eliminates developer maintenance overheads by allowing immediate template modifications, resulting in an estimated 80% decrease in university feedback compilation times.

10. Timeline & Development Milestones

The development phases are structured as follows:

Phase	Milestone Description	Duration
Phase 1	Database setup, schema migration, seed data configurations	Weeks 1 - 2
Phase 2	FastAPI endpoint integrations, authentication rules, backend test suite	Weeks 3 - 4
Phase 3	Hugging Face NLP model caching, keyword search script, query chatbot	Weeks 5 - 6
Phase 4	React layout styling, glassmorphism CSS, charts widgets, responsive forms	Weeks 7 - 8
Phase 5	Form template designer, live testing, user feedback verification, final build	Weeks 9 - 10

11. References

- [1] *FastAPI Documentation*. <https://fastapi.tiangolo.com/> - Web application framework rules and routing definitions.
- [2] *Hugging Face Hub Models*. <https://huggingface.co/models> - Sentiment analysis transformer libraries (DistilBERT).
- [3] *React Documentation*. <https://react.dev/> - Component hierarchies, hook hooks state management.
- [4] *ReportLab User Guide*. <https://www.reportlab.com/docs/reportlab-userguide.pdf> - PDF generation styling.